

PRANAV KAMBLE

Mumbai, Maharashtra

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Education

Veermata Jijabai Technological Institute

B.Tech in Electronics and Telecommunication Engineering (CGPA: 9.34)

Aug 2024 – Present

Mumbai, India

Veermata Jijabai Technological Institute

Diploma in Electronics Engineering (State Rank: 9)

Jun 2021 – May 2024

Mumbai, India

St. Xavier's High School, Airoli

Secondary Education (10th Grade: 89.80%)

2009 – 2021

Navi Mumbai, India

Relevant Coursework

- Analog Electronics
- Digital Electronics
- Power Electronics
- Product Lifecycle (PLM)
- Systems Architecture
- Verilog Programming
- RTL Design
- Machine Learning
- Deep Learning & NLP
- Microcontrollers
- Computer Vision

Experience

Texas Instruments India

Field Applications Engineering (FAE) Intern

May 2026 – Jul 2026

Bangalore, Karnataka

- Architected a **4.6kW residential Battery Energy Storage System (BESS)** by performing market research, topology selection, system architecture design, and TI component selection across power conversion, battery management, isolation, communication, and control subsystems.
- Collaborated with customer engineering teams to understand application requirements, system architectures, and design challenges, recommending appropriate TI analog solutions while supporting design-in and debugging activities.
- Developed technical assets including system architectures, application notes, power electronics simulations, and customer solution proposals through collaboration with multiple TI Business Units.

Projects

Battery Energy Storage System (BESS) | *Power Electronics, System Design, TI Portfolio*

May - Jul 2026

- Designed a **4.6kW residential Battery Energy Storage System (BESS)** featuring a 16S LiFePO₄ battery pack, bidirectional DC-DC converter, Heric inverter topology, and intelligent Battery Management System (BMS).
- Selected TI components across battery monitoring, power conversion, isolation, sensing, communication, and C2000 controller subsystems while optimizing efficiency, safety, and scalability.

TinyNPU – Neural Network Accelerator | *Python, SystemVerilog, VSCode, Icarus Verilog*

Aug 2026

- Built an end-to-end ML-to-hardware inference pipeline for a 21→14→7→1 neural network, implementing FP32 training, backpropagation, quantization, and hardware-ready parameter export.
- Designed SystemVerilog RTL for MAC, ReLU, requantization, neuron, and layer-level datapaths using INT8 multiplication with INT32 accumulation, and verified the complete accelerator hierarchically using Icarus Verilog.
- Developed an INT8/INT32 quantization flow and exported weights, biases, inputs, and reference vectors to `.mem` files for RTL integration and FPGA deployment readiness.

4-bit Mini Processor | *Verilog, RTL Design, Xilinx Vivado*

Jan – Mar 2025

- Designed a custom 4-bit processor implementing a complete Instruction Set Architecture (ISA) with ALU, Register File, Control Unit, Program Counter, and Memory modules in Verilog.
- Verified functionality through Xilinx Vivado simulations and testbenches, strengthening RTL design and digital system integration skills.

A.S.H.A. – AI Symptom & Healthcare Advisor | *FastAPI, ChromaDB, Gemini AI, React*

Jan – Feb 2026

- Developed a multilingual AI-powered healthcare assistant using a structured symptom assessment workflow and Retrieval-Augmented Generation (RAG).
- Integrated ChromaDB with OpenStreetMap APIs to improve response accuracy and enable real-time hospital recommendations.

Technical Skills

Programming: Python, Verilog, C, JavaScript, HTML/CSS, SQL

Hardware: Analog Electronics, Digital Electronics, Power Electronics, RTL Design, Embedded Systems, Battery Energy Storage Systems (BESS)

AI & Software: Machine Learning, Deep Learning, NLP, Computer Vision, RAG, React, Next.js, FastAPI, Node.js

Tools: Xilinx Vivado, TINA-TI, VS Code, Git, GitHub, Jupyter Notebook

Position of Responsibility

Pratibimb VJTI

Chief Operating Officer (COO)

Feb 2026

Mumbai, Maharashtra

- Directed end-to-end operational planning and execution for **Pratibimb**, VJTI's annual cultural festival, coordinating logistics, hospitality, finance, marketing, and event operations for **2,000+ student attendees**.
- Led cross-functional teams and stakeholder communication, driving resource planning, real-time decision-making, and seamless execution throughout the multi-day festival.